



LABORATORY ACTIVITY NO. 1.2.1

Setting Up the Computer

This activity provides a step-by-step instructions for either setting up a new computer or verifying the safe setup of an already existing one. The instructions cover aspects such as reading the PC's manual, positioning the system unit with adequate ventilation, connecting the monitor, mouse, and keyboard, managing cables, and powering on the computer while observing for normal boot signs like LED indicators, fan activity, and hard disk sounds.

I. OBJECTIVES

Upon completion of this laboratory exercise, students are expected to and meet the following intended learning outcomes:

Activity Objectives	Intended Learning Outcomes		
	ILO1	ILO2	ILO3
1. correctly set up a new computer or verify the safe setup of an existing computer by following a step-by-step procedure	•	•	
2. properly connect essential computer peripherals, including the monitor, mouse, and keyboard, to the system unit while managing cables effectively	•	•	•
3. power on a computer system and identify normal operational signs during the boot process, such as power LEDs, fan activity, and hard disk activity.	•	•	

Where:

ILO1 – Exhibit thorough demonstration of the concept

ILO2 – Being able to perform the activity independently

ILO3 – Design and evaluate emerging IT solutions

II. DISCUSSION

- **Module Overview:** This module focuses on "Using a Workstation".
- **Objectives:** The primary objectives are to safely set up a computer system with healthy working practices and to effectively navigate an operating system (OS) and use input devices.
- **Computer Setup:**
 - Instructions for setting up a personal computer include checking contents, reading manuals, and positioning devices and cables.
 - Emphasis is placed on proper placement to avoid trip hazards, ensure adequate airflow (e.g., 6" space around the unit), and maintain usability.



- Steps involve connecting peripherals (monitor, mouse, keyboard) and power cables.
- Cable management is advised to keep them tidy and prevent trailing under desks or across walkways.
- The process of powering on includes pressing monitor and PC power switches and observing normal boot signs like power LEDs, spinning fans, and hard disk activity (listening for healthy sounds, not grinding or clicking).
- **Ergonomics:** Important ergonomic concepts include proper keyboard and mouse placement, sitting position, and monitor placement to prevent Repetitive Strain Injury (RSI).
- **OS Navigation:**
 - Covers signing in to the computer, launching software applications, managing data files, and configuring the system.
 - Detailed steps for "Signing In to Windows" involve understanding the computer boot process/POST, sign-in IDs, authentication, usernames, passwords/credentials, passing the lock screen, selecting user accounts, and entering passwords.
- **Desktop and Taskbar Usage:** Explains features like desktop shortcuts, the Start/Windows button, pinned icons, the notification area, and the Start Screen/Start Menu.
- **Start Screen Interaction:** Describes the Start Screen as the main interface for Windows 10, how to access it via the Start key, its function for mobile hardware, and its use of tiles, apps, and instant search.
- **Customization:** Instructions are provided for customizing the Start Screen and Taskbar, including pinning apps, grouping tiles, and adjusting personalization settings.
- **Input Devices:**
 - **Mouse/Touchpad:** Covers selecting/opening icons, selecting menu commands, using shortcut menus, drag and drop, and scrolling.
 - **Keyboard:** Discusses typing ordinary text, touch typing finger positions, keyboard layouts, and the function of special/command keys (e.g., ENTER/RETURN, CTRL, ALT, TAB, ARROW keys, SHIFT, NUM LOCK, START/Windows key, SHORTCUT key, FN keys, BACKSPACE, DELETE).
 - **Touchscreen:** Explains gesture support (tap, tap and hold, pinch and stretch, swipe) and the use of an on-screen keyboard.
- **Desktop Icons:** Teaches recognition of various desktop icons such as files, shortcuts, folders, applications, and devices.
- **Working with Windows:** Includes a visual example of interacting with Windows, likely through File Explorer.



III. MATERIALS

A Windows 11 installed computer system.

IV. PROCEDURE

For this activity, you will execute the provided step-by-step instructions to either correctly set up a new computer or verify the safe configuration of an existing computer. As you complete each step, capture a screenshot of the resulting output or completed action that directly demonstrates successful execution of that instruction. Paste each screenshot into the designated area within your lab document.

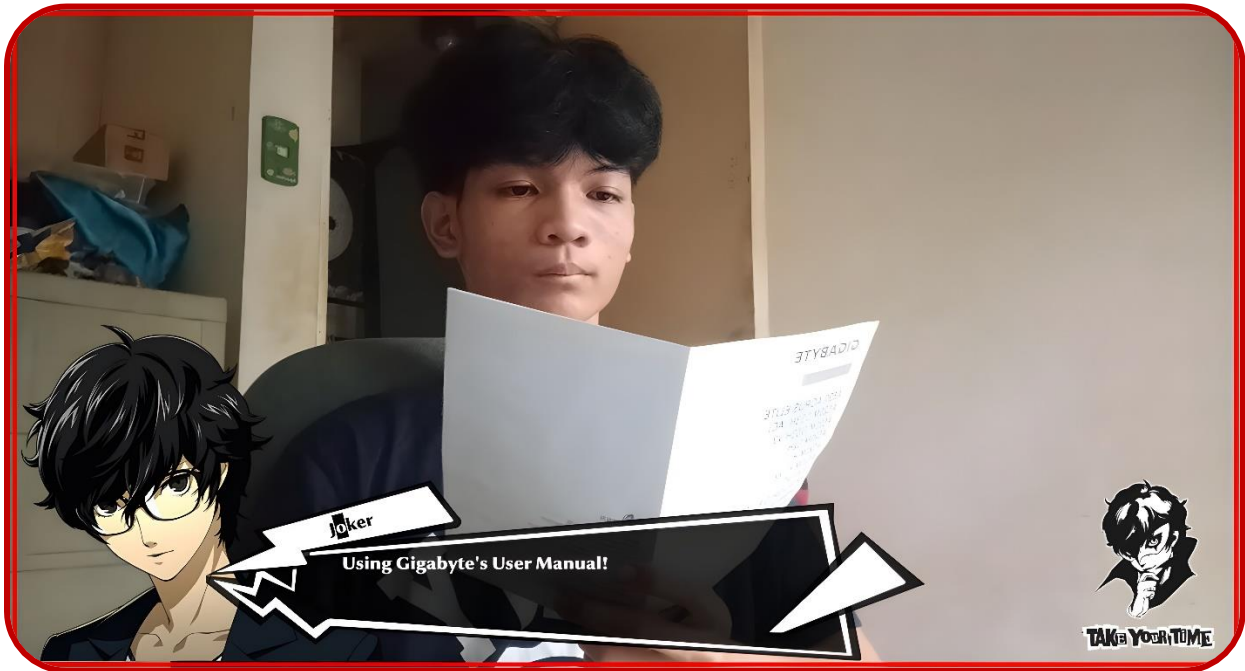
Important Considerations:

- **New Computer Setup:** If you are working with a new computer, meticulously follow all steps to ensure a proper and safe initial setup.
- **Existing Computer Verification:** If you are using an already configured computer, perform each step to meticulously check that it is set up safely and according to the best practices outlined.
- **Unperformable Steps:** If you are unable to physically perform a specific instruction due to hardware limitations, lack of access, or other constraints, you must search the internet for a relevant, high-quality image that accurately depicts the intended outcome or action of that instruction.
- **Image Citation:** When using an image from the internet, you must include a clear citation for the source of the image immediately below it. This ensures academic integrity and gives credit to the original creator.

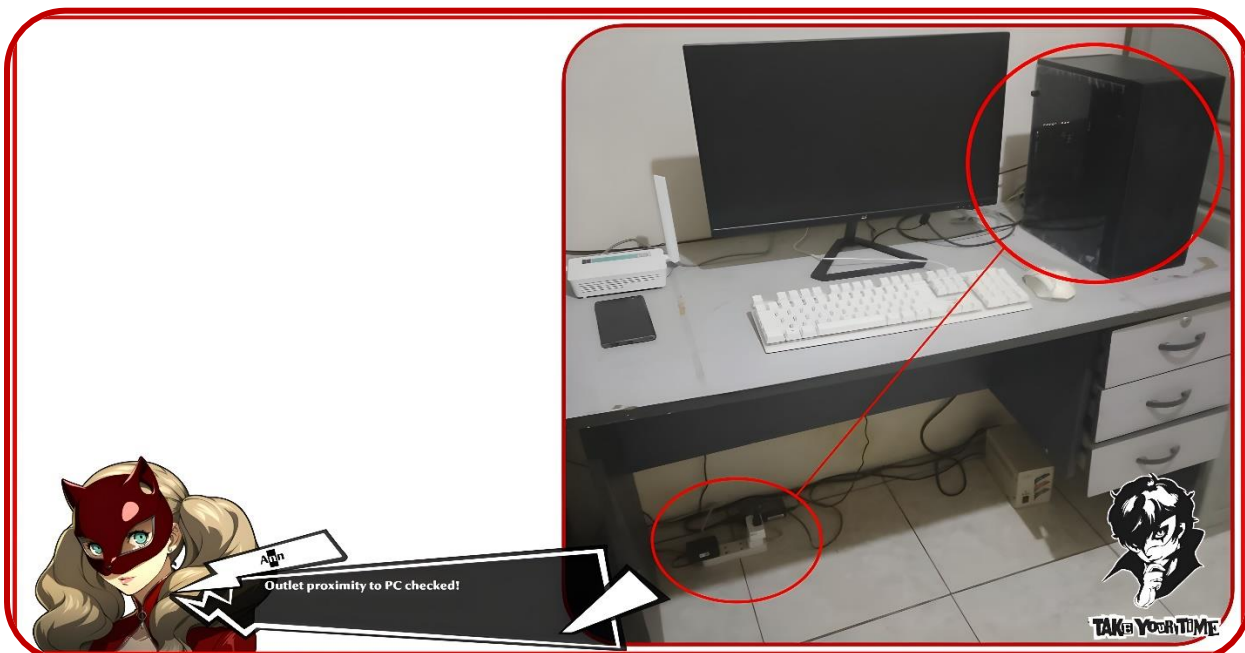


Steps:

- 1) Read the PC's instruction manual and make sure the contents listed on the box are all present.



- 2) Put the system unit on the desk or floor - remember the following points:
 - a) Close enough to an electrical outlet





- b) Close enough to your desk so that cables for the mouse and keyboard are not stretched.



- c) 6" space around the unit for air to circulate freely.

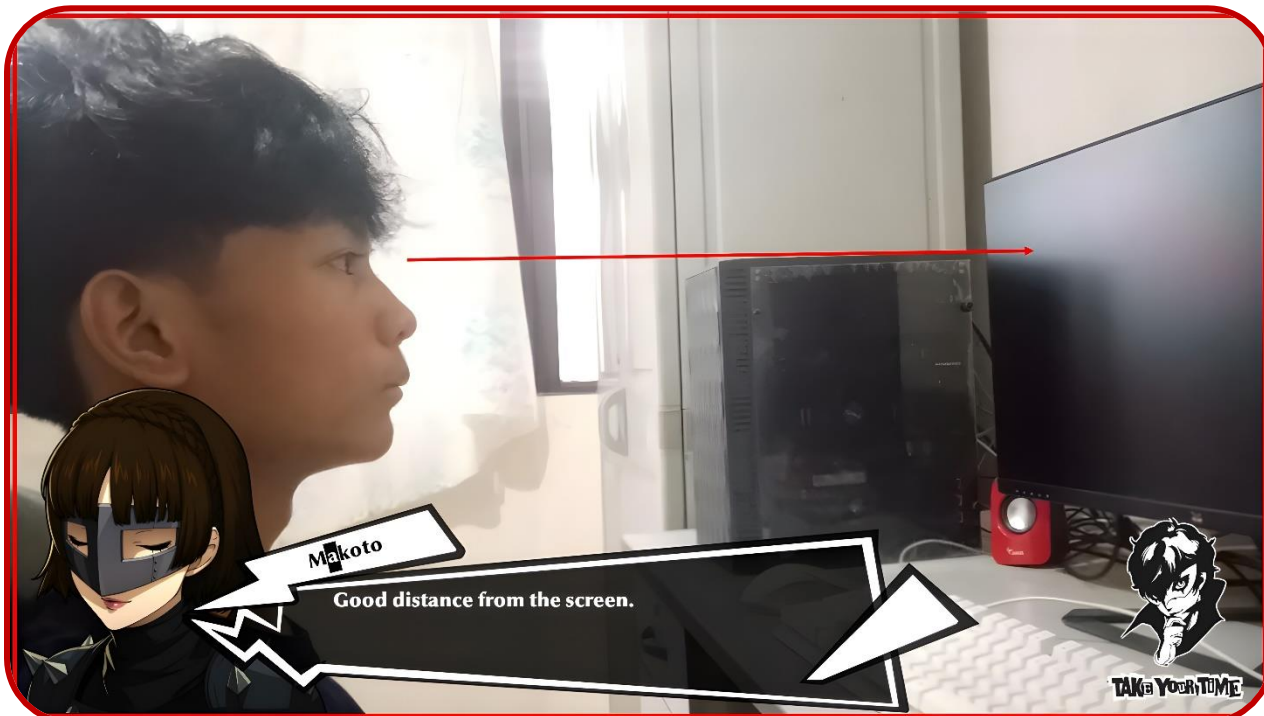




- d) Not positioned close to a radiator and ideally not exposed to direct sunlight.

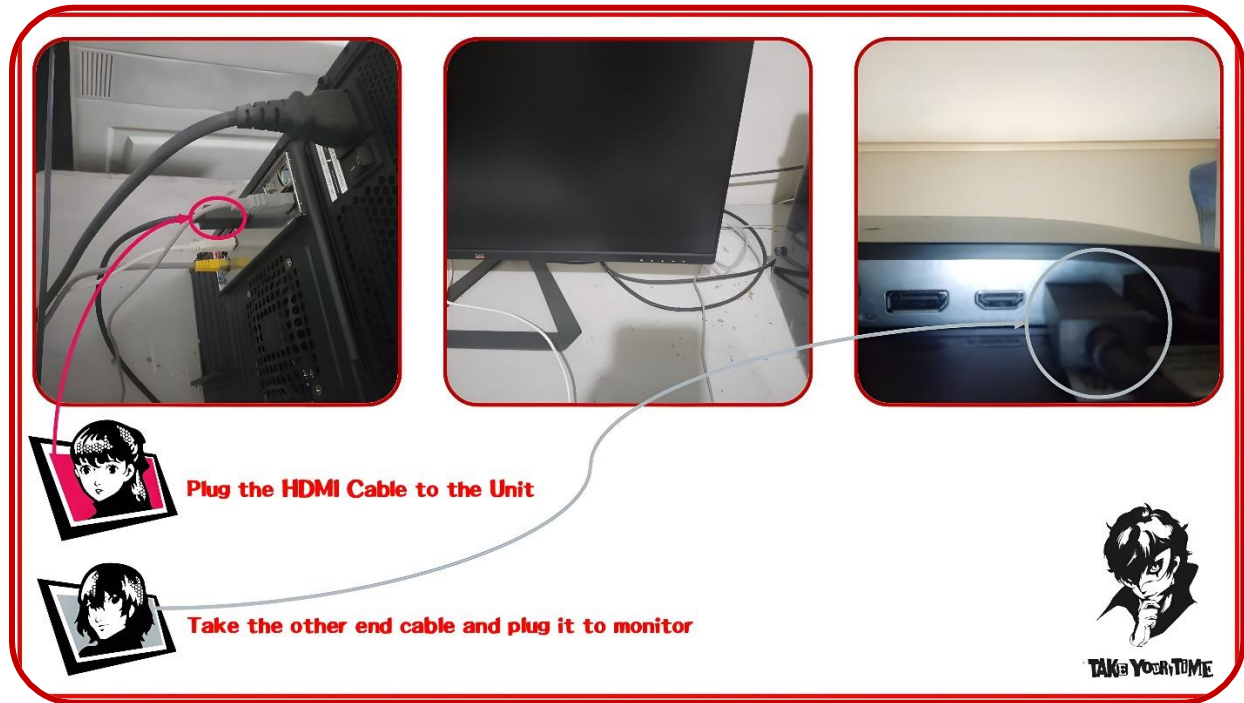


- 3) Position the monitor on your desk and adjust it and your chair so that the top of the screen is level with your eyes.

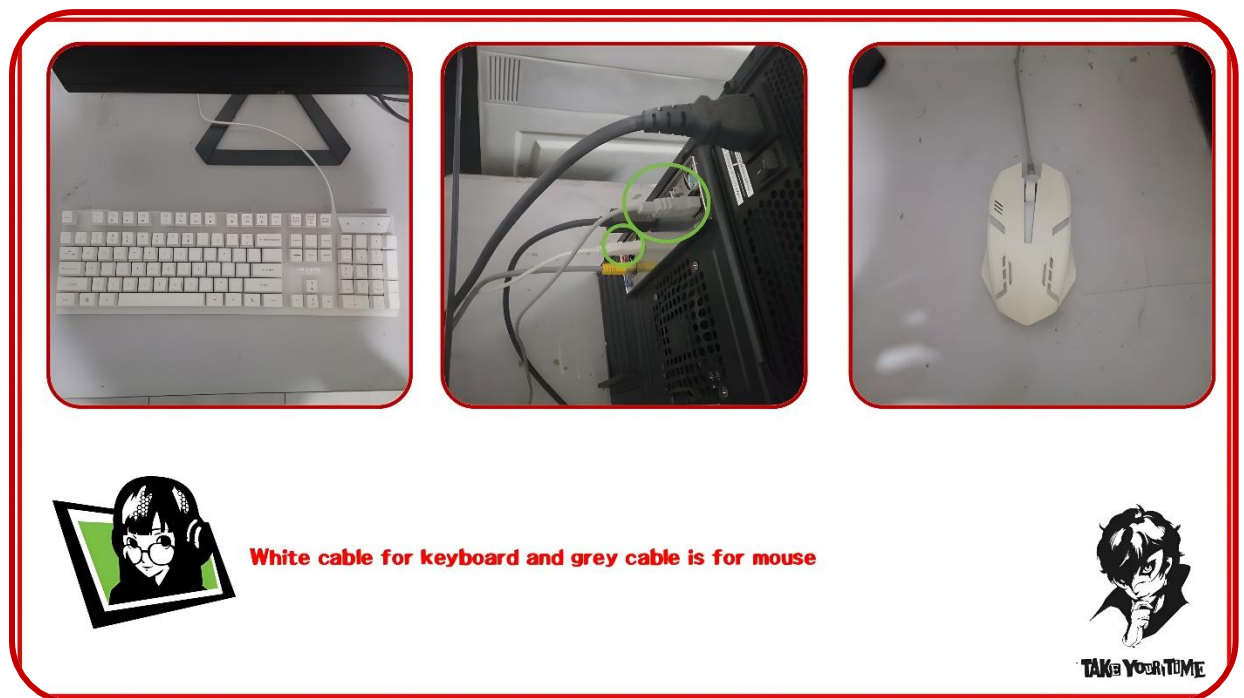




- 4) Connect the monitor to the port on the system unit, using the connector as shown in your system documentation.

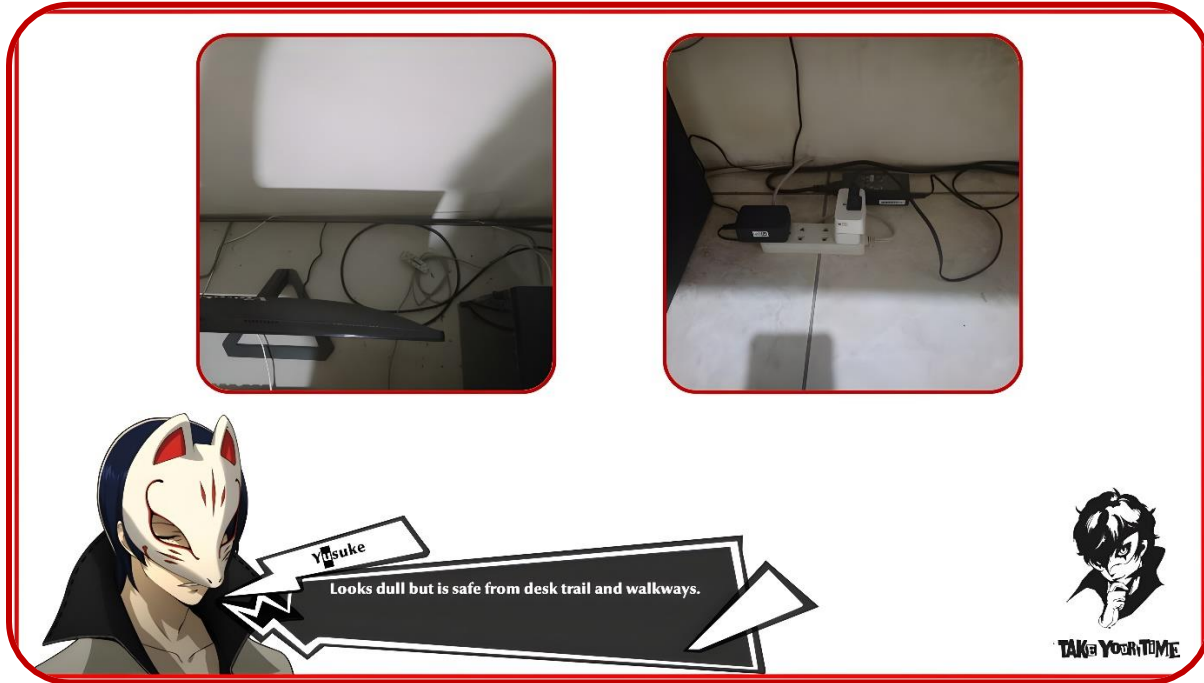


- 5) Connect the mouse and keyboard to the ports on the system unit, using the connector as shown in your system documentation.

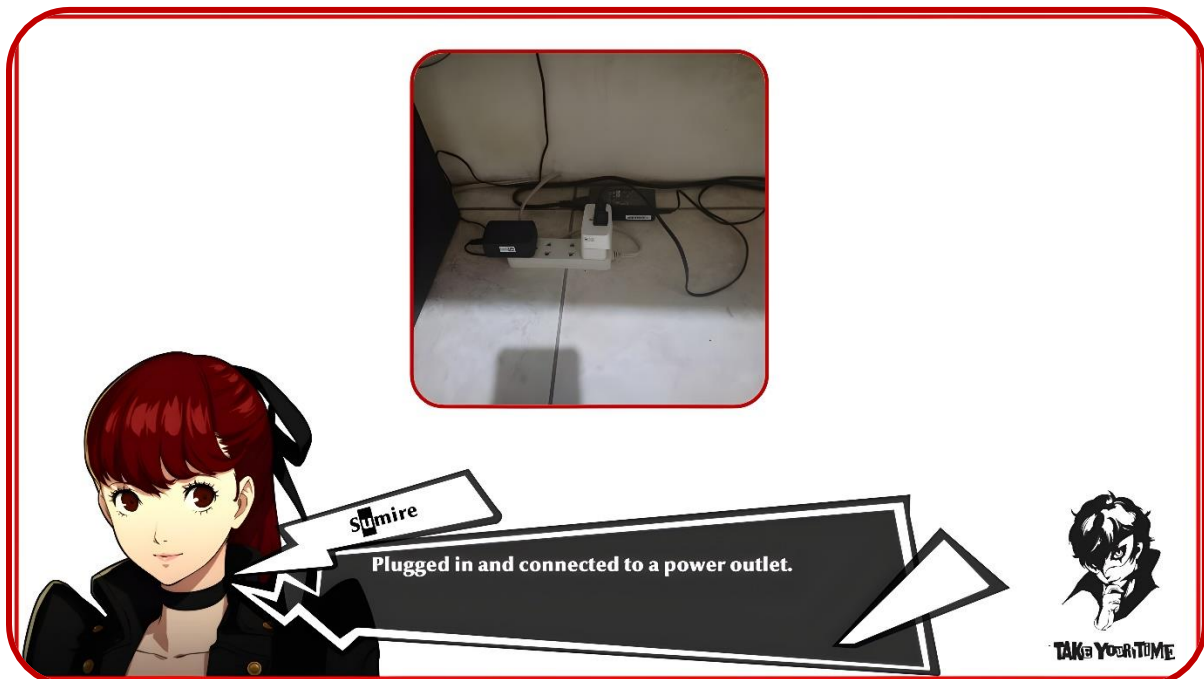




- 6) Try to keep the cables tidy so that they do not trail under your desk or cross walkways.



- 7) Plug the monitor into a building power outlet or into the power socket on the system unit (check the system documentation to find out how the monitor is powered).





- 8) Plug the power cable into the back of the system unit and then into the building power. Turn on the building power wall switch.

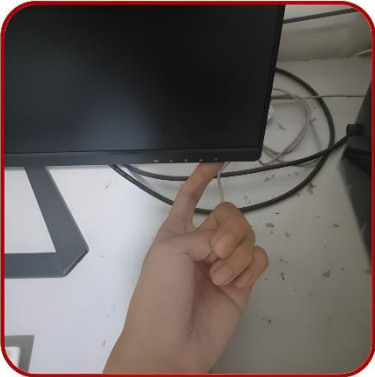


Sumire
Plugged in and connected to a power outlet.



TAKAHASHI

- 9) Press the power switch on the monitor then the power switch on the PC.



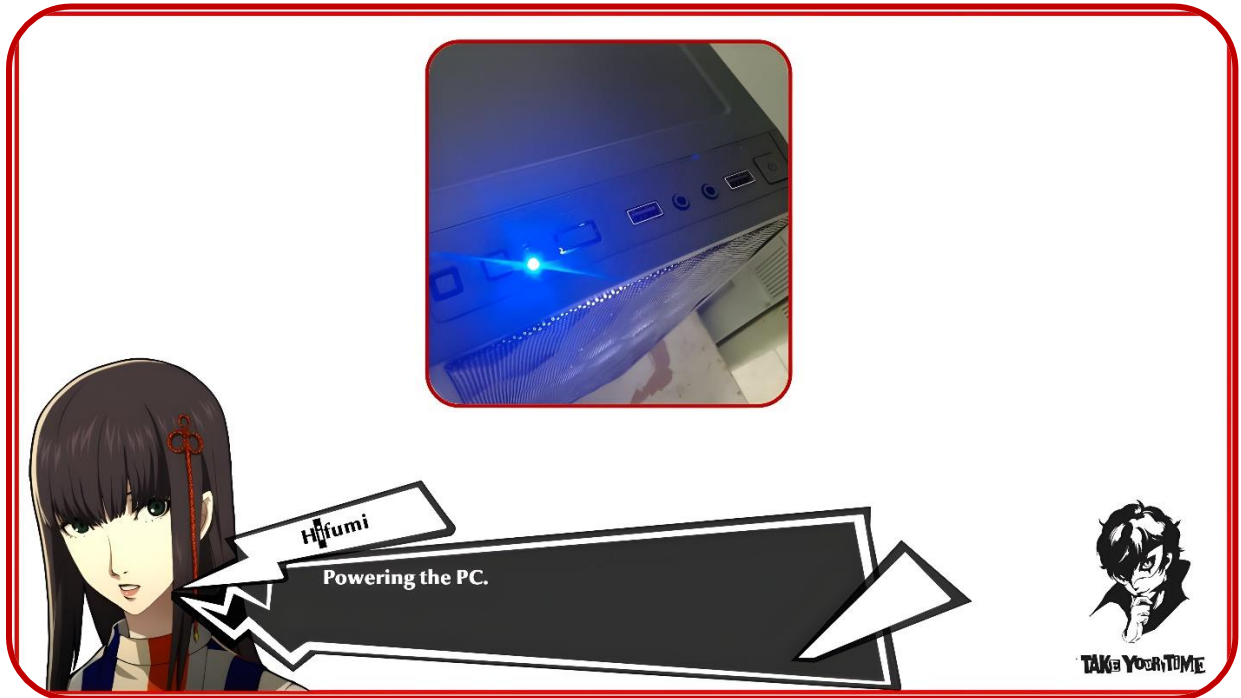
Hifumi
Powering the PC.



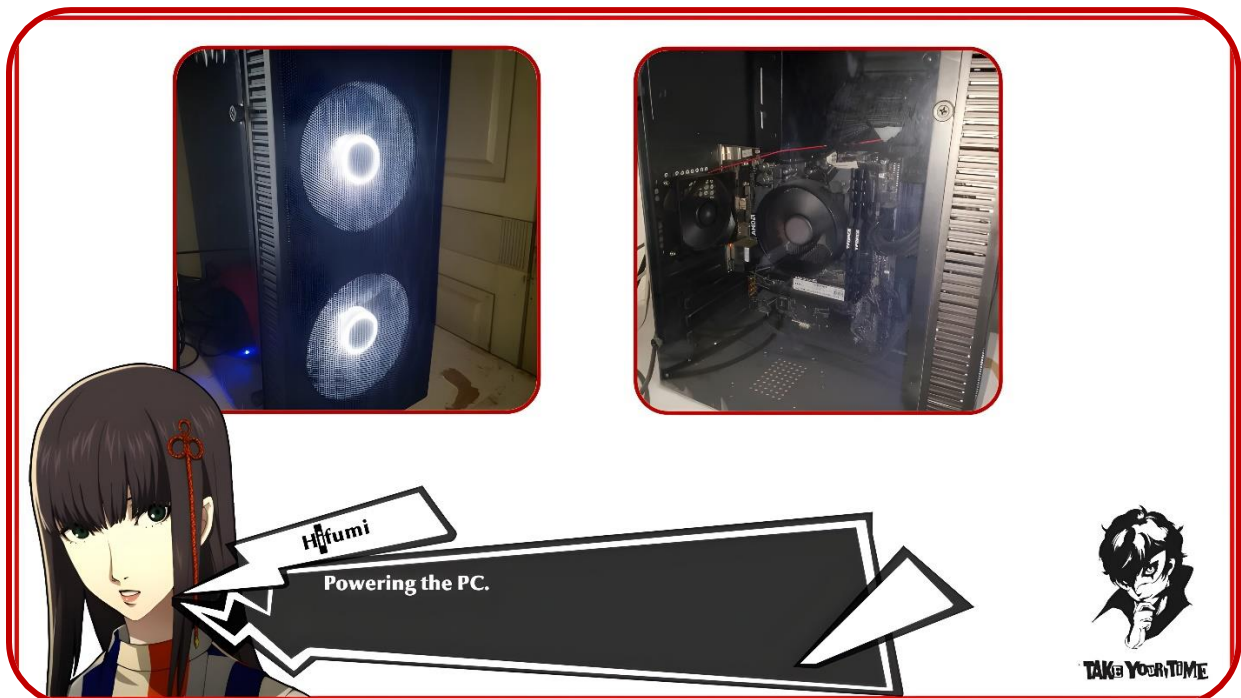
TAKAHASHI



- 10) Wait for the computer to start ("boot"). As the PC powers up, listen and look for the following signs that the computer is operating normally
- Power LED comes on

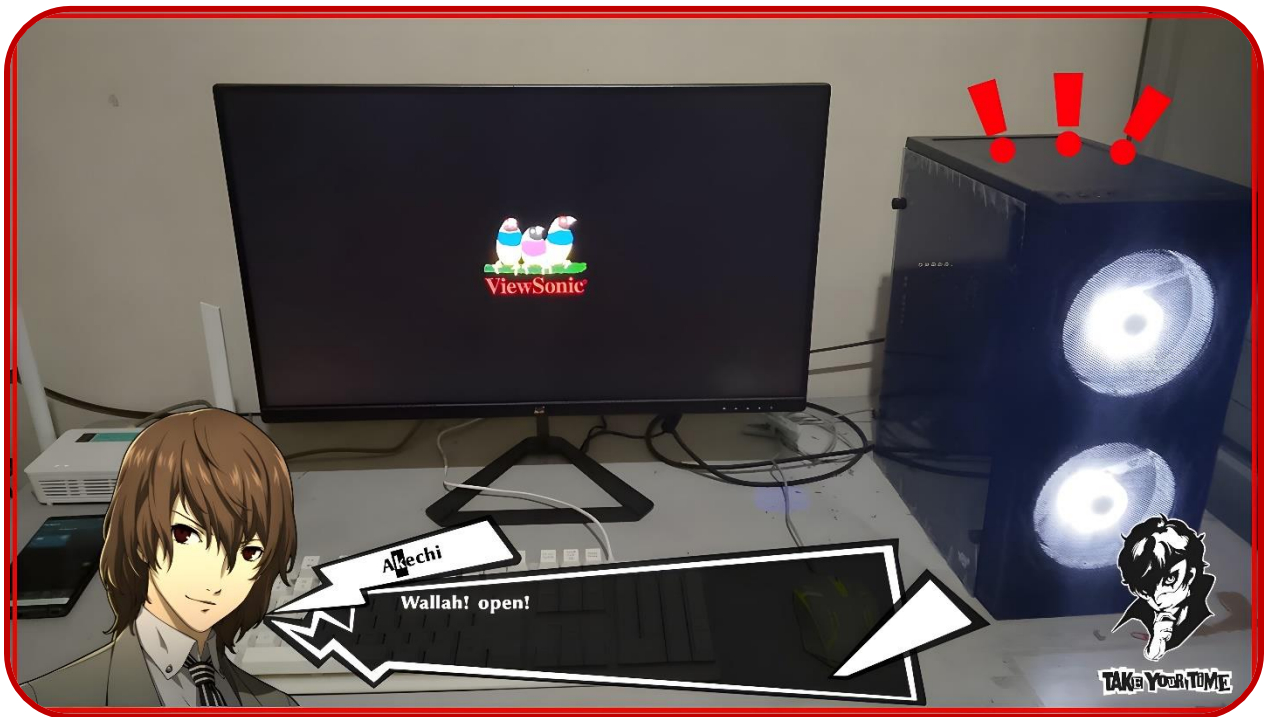


- Fans start spinning

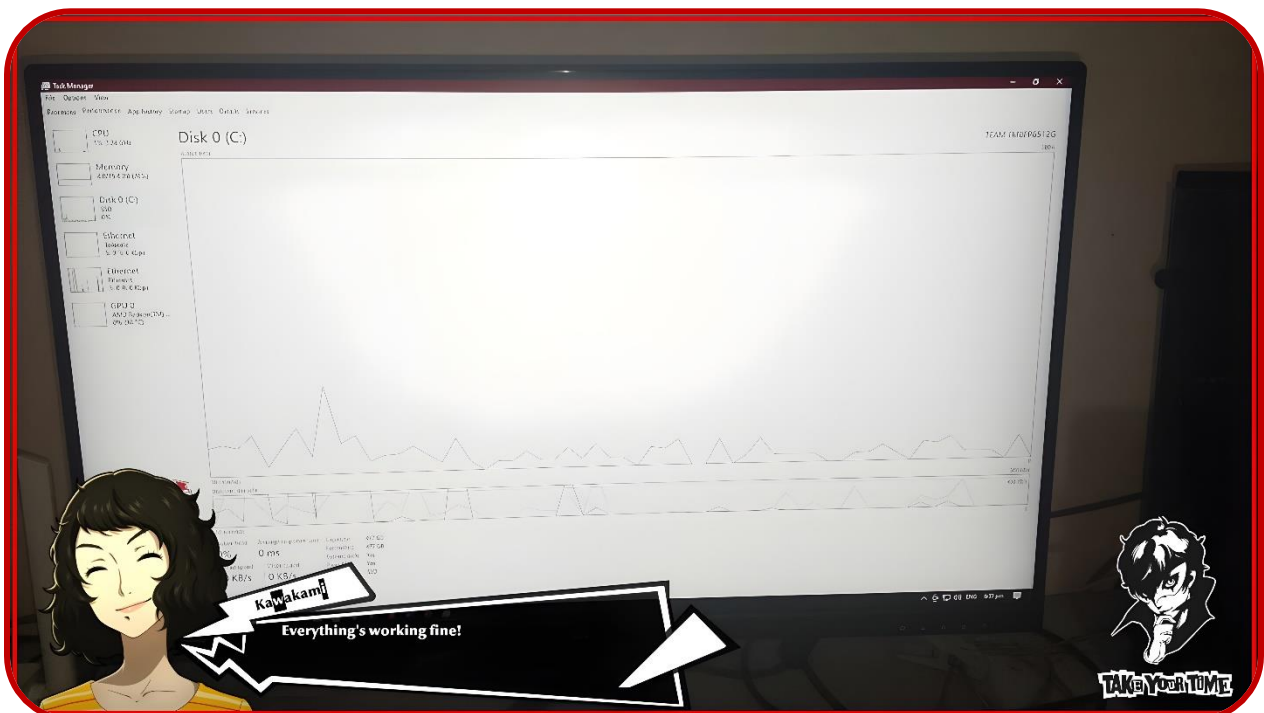




- c. Single short beep (most computers these days do not beep though)



- d. Hard disk activity (LED flickers and there may be some soft noise [grinding or clicking indicates a problem])





V. REFLECTION:

Upon successful completion of "Lab 1.2.1: Setting up the Computer," write a comprehensive reflection (between 150-250 words) addressing the following points:

- **Learning Outcomes:** What new knowledge or skills did you acquire or reinforce regarding computer setup and basic operation?
- **Challenges and Solutions:** Describe any difficulties or unexpected issues you encountered during the lab. How did you overcome them, or what steps would you take to resolve them in the future?
- **Real-world Relevance:** How do the skills and understanding gained in this lab relate to real-world scenarios, either in your personal life or a professional IT context?
- **Critical Thinking:** If you had to use an internet image for any step, explain why you couldn't perform the action physically and how finding the image contributed to your understanding.
- **Suggestions for Improvement:** What suggestions do you have to improve this lab activity for future students?

Completing this activity helped me gain a better understanding of the proper way to assemble and prepare a computer for use. I reinforced my knowledge of identifying computer hardware, connecting peripherals such as the keyboard, and mouse, organizing cables properly, and following the correct startup procedures. These skills are fundamental for me and anyone pursuing a career in IT because they do form the foundation for advanced computer maintenance and troubleshooting. One challenge I encountered during the activity was my cables are very messy. What I did was I unplugged everything and tried my best to do separations to make it somehow detectable or able to know which cables go and not look like spaghetti. I did not use any images online because its all provided for me. The knowledge and skills I gained from this are valuable in both personal and professional situations. They can be applied when setting up a home computer, helping others troubleshoot basic hardware issues, or working in IT support. To improve this for future students, I suggest providing more computer activities, allowing additional hands-on practice time (which is I do learn better), and including simple troubleshooting exercises that simulate common real-world computer setup problems.